

ECE 461 - HW 11

Buran Aybak Kilic

- 1) Slow, flat Rayleigh fading channel $y[m] = h x[m] + w[m]$ where $x[m] = x^T[m] \in \{-\sqrt{E}, -\frac{\sqrt{E}}{3}, \frac{\sqrt{E}}{3}, \sqrt{E}\}$
 $\hookrightarrow h^I, h^Q \sim \mathcal{N}(0, A/2)$ indep. $\hookrightarrow w^I[m], w^Q[m] \sim \mathcal{N}(0, \sigma^2/2)$ indep

$$y[m] = h x[m] + w[m] \begin{cases} \text{Re}\{y[m]\} = h^I x[m] + w^I[m] \\ \text{Im}\{y[m]\} = h^Q x[m] + w^Q[m] \end{cases} \left. \begin{array}{l} \text{transmitted separately over in-phase and} \\ \text{quadrature channels} \end{array} \right\} x[m] \in \mathbb{R}$$

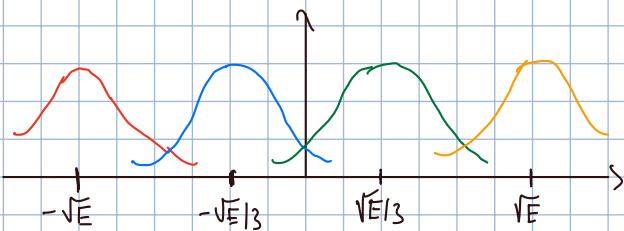
Using a matched filter in receiver: $\tilde{y}[m] = h^I \text{Re}\{y[m]\} + h^Q \text{Im}\{y[m]\}$

$$= \underbrace{[(h^I)^2 + (h^Q)^2]}_{\tilde{h}} x[m] + \underbrace{h^I w^I[m] + h^Q w^Q[m]}_{\tilde{w}[m]}$$

$$\rightarrow \text{Var}[\tilde{w}[m]] = [(h^I)^2 + (h^Q)^2] \frac{\sigma^2}{2} = \frac{|h|^2 \sigma^2}{2} \Rightarrow \tilde{w}[m] \sim \mathcal{N}(0, \frac{|h|^2 \sigma^2}{2}), \tilde{h} = |h|^2$$

$$\tilde{y}[m] = |h|^2 x[m] + \tilde{w}[m] \Rightarrow \text{effectively an AWGN channel}$$

$$\tilde{\text{SNR}} = \frac{|h|^4 E}{|h|^2 \sigma^2/2} = 2 |h|^2 \text{SNR}$$



$$P_e = \frac{1}{4} \left[\underbrace{2 \cdot 2Q\left(\sqrt{\frac{|h|^4 (E/9)}{|h|^2 \sigma^2/2}}\right)}_{\text{errors from } \pm \sqrt{E}/3} + \underbrace{2 \cdot 1Q\left(\sqrt{\frac{|h|^4 (E/9)}{|h|^2 \sigma^2/2}}\right)}_{\text{errors from } \pm \sqrt{E}} \right]$$

$$P_e = \frac{3}{2} Q\left(\sqrt{\frac{2}{9} |h|^2 \text{SNR}}\right)$$

* Since $h^I, h^Q \sim \mathcal{N}(0, A/2)$, $|h|^2 \sim \text{Exp}(1/A)$

* In class we've derived that $\mathbb{E}_{|h|^2} [Q(\sqrt{2 |h|^2 \text{SNR}})] = \frac{1}{2} \left(1 - \sqrt{\frac{A \cdot \text{SNR}}{A \cdot \text{SNR} + 1}} \right)$

$\hookrightarrow$ Note that we have to take expected value since channel is not constant $\Rightarrow$ probabilistic

$$\text{Then } \mathbb{E}_{|h|^2} [Q(\sqrt{\frac{2}{9} |h|^2 \text{SNR}})] = \frac{1}{2} \left[1 - \sqrt{\frac{A \cdot \text{SNR}/9}{A \cdot \text{SNR}/9 + 1}} \right]$$

$\Rightarrow$

$$\tilde{P}_e = \mathbb{E}[P_e] = \frac{3}{4} \left(1 - \sqrt{\frac{A \cdot \text{SNR}}{A \cdot \text{SNR} + 9}} \right)$$

At high SNR, we can approximate the term in square-root via Taylor Series

$$\sqrt{\frac{A \cdot \text{SNR}}{A \cdot \text{SNR} + 9}} = \left(\frac{A \cdot \text{SNR} + 9}{A \cdot \text{SNR}} \right)^{-1/2} = \left(1 + \frac{9}{A \cdot \text{SNR}} \right)^{-1/2}$$

Noting that $(1+x)^{-1/2} \approx 1 - \frac{1}{2}x \longrightarrow \sqrt{\frac{A \cdot \text{SNR}}{A \cdot \text{SNR} + 9}} \approx \left(1 - \frac{9}{2 \cdot A \cdot \text{SNR}} \right)$

$$\tilde{P}_e = \frac{3}{4} \left[1 - \left(1 - \frac{9}{2 \cdot A \cdot \text{SNR}} \right) \right] = \boxed{\frac{27}{8 \cdot A \cdot \text{SNR}}} \quad \text{in high SNR}$$

2) For Rayleigh fading channel $P\{\text{outage}\} = P\{|h|^2 < \frac{q_{th}}{\text{SNR}}\} = 1 - e^{-q_{th}/A \cdot \text{SNR}} \approx \frac{q_{th}}{A \cdot \text{SNR}}$

→ Consider 3 i.i.d. Rayleigh channels with $|h_1|^2, |h_2|^2, |h_3|^2 \sim \text{Exp}(1/A)$ which are independent

$$P\{\text{outage}\} = P\{\max\{|h_1|^2, |h_2|^2, |h_3|^2\} < \frac{q_{th}}{A \cdot \text{SNR}}\}$$

* Notice that if max of 3 random variables is smaller than $\frac{q_{th}}{\text{SNR}}$, then all 3 random variables are separately smaller than q_{th}/SNR

$$P\{\text{outage}\} = P\{|h_1|^2 < \frac{q_{th}}{\text{SNR}}, |h_2|^2 < \frac{q_{th}}{\text{SNR}}, |h_3|^2 < \frac{q_{th}}{\text{SNR}}\}$$

Since $|h_1|^2, |h_2|^2, |h_3|^2$ are indep. and identically distributed.

$$P\{\text{outage}\} = P\{|h_1|^2 < \frac{q_{th}}{\text{SNR}}\} \cdot P\{|h_2|^2 < \frac{q_{th}}{\text{SNR}}\} \cdot P\{|h_3|^2 < \frac{q_{th}}{\text{SNR}}\} = \left[P\{|h_1|^2 < \frac{q_{th}}{\text{SNR}}\} \right]^3$$

$$P\{|h_1|^2 < \frac{q_{th}}{\text{SNR}}\} = \int_0^{q_{th}/\text{SNR}} \frac{1}{A} e^{-x/A} dx = 1 - e^{-\frac{q_{th}}{A \cdot \text{SNR}}} \implies$$

$$\boxed{P\{\text{outage}\} = \left[1 - e^{-q_{th}/A \cdot \text{SNR}} \right]^3}$$

In high SNR $\Rightarrow e^{-q_{th}/A \cdot \text{SNR}} \approx 1 - q_{th}/A \cdot \text{SNR}$

$$\boxed{P\{\text{outage}\} = \left[\frac{q_{th}}{A \cdot \text{SNR}} \right]^3}$$